

[affiliation not assigned]

August 3, 2026

The Editors
Physical Review D

Dear Editors,

We submit for your consideration the manuscript “*Symmetry control of exceptional structure in the massive scalar quasinormal spectrum of doubly rotating five-dimensional Myers–Perry black holes.*”

Exceptional points have become an active topic in black-hole spectroscopy, and exceptional lines have recently been established in the de Sitter members of the Myers–Perry family. It is natural to expect that the asymptotically flat, doubly rotating case should behave similarly, and in particular that the enhanced $U(2)$ symmetry at equal angular momenta should generate degeneracies which unfold into exceptional points once the symmetry is broken. That is the standard mechanism for symmetry-protected exceptional structure in non-Hermitian systems.

Our central point is that this expectation fails, and fails for an exact reason. Because both azimuthal Killing vectors survive for arbitrary rotation parameters, the quasinormal pencil is a direct sum over azimuthal sectors, and the Jordan chains of a direct sum are the union of the summands’ chains. A coincidence of frequencies from different sectors is therefore always semisimple. The equal-spin multiplet degeneracies are protected from splitting and forbidden from being defective by one and the same fact. We regard this structural statement, rather than the accompanying numerics, as the main contribution.

The remainder of the paper takes the redirection seriously. We classify the discrete symmetry group that survives — a Klein four-group acting on parameters *and* sector labels — and derive from it a termination law: in diagonal sectors an exceptional line, if one existed, would have to occur in mirror pairs and terminate quadratically on the equal-spin surface rather than crossing it. We then search the one channel the no-go leaves open, with a branch atlas of 50 558 continued points across seven sectors, and report a bounded exclusion with three rescaling-invariant diagnostics.

Two methodological points may interest referees independently of the physics. First, we show that the derivative of a spectral condition cannot bound the distance to an exceptional point, because the condition is defined only up to a nonvanishing analytic factor; we give an invariant replacement and exhibit two conditions with identical zeros whose derivatives differ by more than 10^5 . Second, we report two branch-tracking defects that each produced a spurious near-degeneracy, and the diagnostics that caught them. We include these because we believe the failure modes are generic to this class of search.

We are explicit about what is not established. The exclusion is bounded, over a declared domain, and is not a theorem; its minimum is attained on the boundary of the searched box; and the quasiresonant regime is not searched, because there the recurrence-free solvers are inapplicable by formulation rather than by convergence failure — a point we establish quantitatively and which we believe corrects a common misattribution of such failures to near-extremality.

All code, machine-readable results and reproduction scripts are publicly available, and the central

result is regenerated by a single command from a clean clone.

We suggest as possible referees **[referee suggestions not assigned]**.

Sincerely,

[author names not assigned]